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DAILY EDITION

*Benjamin you stop pickin' the
bark off of that tree!*

— GRANDMA BRANDT

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Speaking at New York C++ meetup on January 13

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Math

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Third SAIR competition: inverse Galois challenge

[The third SAIR challenge has been launched, focusing on generating numerical data for the inverse Galois problem. This collaborative project with the L-functions and modular forms database (LMFDB) aims to complete a large, verified mathematical dataset, with verification performed by MAGMA and PARI/GP.]

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Why the Human Genome's Tangled Physicality May Confound AI

[Despite the human genome's full sequencing, its intricate physicality and the vast non-coding regions that regulate gene activity present a complex puzzle, challenging the initial view of DNA as a simple blueprint. Understanding how genes are switched on and off, rather than just their sequence, is now the critical question, potentially confounding AI's ability to fully decipher life's secrets.]

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Epson HX-20 Gets A Drive Upgrade

[Andrew Menadue has ingeniously upgraded the vintage Epson HX-20, replacing its unreliable microcassette drive with a modern solution. Utilizing a Raspberry Pi Pico and microSD card, his device emulates the original tape drive and can even mimic ROM cartridges, ensuring the classic computer's longevity.]

Space

SPACE · [spacenews.com] · [today] · 4,474 tokens / 0.0714c

The MEO durability crisis: why LEO hardware will fail the new orbital economy

The space industry faces a "MEO durability crisis" as it attempts to use low Earth orbit (LEO) hardware, designed for short-term use and limited radiation shielding, in the much harsher medium Earth orbit (MEO). This systemic issue, extending beyond electronics to structural composites, demands a radical rethinking of material durability for permanent orbital infrastructure.

To some, space is the final frontier. But the reality is that once you leave Earth's atmosphere, you're not entering a single, uniform void. There are several frontiers up there — distinct orbital zones — and each throws a very different set of problems at us. What works perfectly in one orbit might fail completely in another. As the space industry accelerates toward a fast-growing multi-orbit economy in 2026, we're bringing our low Earth orbit (LEO) habits with us. LEO hardware is fantastic for what it does. It's characterized by high-volume manufacturing and cost-effective components, including limited radiation shielding. But dragging LEO technology into medium Earth orbit (MEO), a harsher zone sitting between 2,000 and 36,000 kilometers above us, is creating a massive MEO durability crisis. MEO demands another level of environmental protection and adaptation that standard commercial-off-the-shelf electronics and other components simply aren't capable of. While the vulnerability of these electronics is well-documented, it's really just the blinking "check engine" light of a much deeper, systemic issue. Beneath the avionics and shiny solar arrays lies a massive, under-discussed materials science crisis. We are essentially trying to build a more permanent orbital infrastructure using materials designed for short-term use. If we don't radically rethink our approach to material durability — specifically regarding the structural composites that form the backbone of modern or-

bital vehicles — our grand cislunar architecture will physically degrade before it can mature.

The "Stay and Serve" paradigm

Historically, our operational footprint beyond LEO has been defined by short-term "launch and burn" missions. Upper stages, kick motors and transfer vehicles do their job, fire their thrusters and then either retire to graveyard orbits or burn up on reentry. But the emerging orbital economy isn't short-term use, and that presents a very different set of challenges. The next decade will be defined by Orbital Transfer Vehicles (OTVs), orbital gas stations and bustling satellite servicing hubs. These satellites aren't transient tourists. They are expected to operate in MEO and geosynchronous equatorial orbit (GEO) for years, repeatedly docking with client satellites, pumping potentially volatile cryogenic propellants and maneuvering assets into new positions. This operational shift introduces a whole new world of mechanical stress. Standard LEO hardware simply doesn't have the structural stamina for a multi-year "stay and serve" lifestyle. Imagine the continuous, cyclic loading of repeated docking operations, mixed with the wild temperature swings of MEO. Every time a servicing vehicle catches a client satellite, a physical shockwave ripples through the chassis and those highly pressurized fuel tanks. Over years of service, these constant bumps and bruises risk pushing standard structural materials well past their fatigue thresholds. While commer-

cial operators rarely publicize hardware failures, NASA provided proof that LEO baselines will fail in MEO. When designing the Van Allen Probes, engineers found that standard, high-heritage LEO design practices were insufficient. To survive the harsh radiation belts, NASA had to abandon commercial-off-the-shelf components in favor of a heavily customized architecture featuring extensive structural shielding, radiation-hardened electronics and specialized fault-management software. Plus, the probes were built for a seven-year mission. Today's commercial MEO assets are being tasked with 15-year lifespans. Expecting LEO hardware to double the lifespan of specialized MEO satellites isn't just optimistic — it's a multi-billion-dollar gamble against physics.

The unsung hero under siege

To truly understand this durability crisis, we have to zoom in on the unsung hero of spacecraft structures and pressure vessels: the epoxy resin. Modern spaceflight loves carbon fiber composites, and for good reason — they offer incredible strength and durability without the heavy payload penalty. In a composite pressure vessel, the carbon fibers are the muscle, providing crucial tensile strength. But it's the intricate chemical lattice of the epoxy resin that acts as the glue, holding the entire matrix together. That is, until they reach MEO and face the punishing environment of the higher-energy Outer Van Allen radiation belts. Here, spacecraft are subjected to intense ionizing radiation, prolonged exposure to the vacuum of space, and extreme thermal cycling. This environment attacks the material on two distinct fronts:

- **Radiation Damage:** High-energy radiation can degrade the mechanical properties of specific carbon fiber materials and break down the essential polymeric bonds within the resin systems.
- **Outgassing:** Under vacuum conditions and intense thermal cycling, standard resins suffer from outgassing, releasing volatile organic compounds and moisture that can migrate away from the material.

This is a double-barrelled threat to the commercial space economy: Severe operational issues: First, those evaporated compounds don't just disappear;

they adhere to cold surfaces. They condense on sensitive optics, star trackers and camera lenses — not just on servicing vehicles, but on the multimillion-dollar client satellites they are sent to refuel. These outgassed molecules can also condense on solar panels, which are critical to meeting the mission's energy requirements. Shattering the matrix: Second is the effect on the tank itself. As the resin loses its chemical integrity through radiation and outgassing, the structural lattice is left dangerously compromised. The once-resilient polymer matrix turns brittle. Micro-cracks start to creep through the structure, and the very tanks meant to safely store highly pressurised propellants become vulnerable to catastrophic structural failure.

The benefits of pre-preg and the chemical lattice

The reality is that we can't simply "scale up" LEO manufacturing techniques to conquer MEO and cislunar space. Adding thicker walls to composite tanks to compensate for degradation only cannibalizes payload mass, defeating the very economic purpose of using composites in the first place. Instead, the solution lies in chemistry. So what are the solutions? Re-engineering the chemical lattice of our composites to prevent long-term fatigue is fundamental to the growing MEO industry's efforts to create a safe, commercial reality. The aerospace sector must prioritize the development and rigorous qualification of novel, radiation-hardened resin systems that resist polymer degradation. We need matrix materials engineered at the molecular level to withstand deep-space outgassing and the cumulative fatigue of repeated docking maneuvers, all without sacrificing the lightweight properties that make orbital logistics economically viable. There is amazing work being done with NASA-backed polybenzoxazines and cyanate esters, but these are often prohibitively expensive and require high-temperature curing processes. Additionally, adapting our manufacturing approaches can bridge this gap today. For example, transitioning from traditional wet winding to pre-preg composite fibers — where filaments are pre-impregnated with specialized polymers under strictly controlled laboratory conditions — delivers a uniform consistency that

wet winding cannot match. This enables a thinner, highly uniform, and stronger overwrap for Composite Overwrapped Pressure Vessels better suited for the rigors of MEO. The next phase of the “new orbital economy” requires shifting these advanced manufacturing paradigms from expensive, bespoke deep-space probes into high-volume, commercial production. As an industry, our focus has overwhelmingly been on the software, sensors and launch vehicles required to reach higher orbits. But reaching MEO is only half of the journey; surviving there is the true test. The launch-and-burn materials of the past will not sustain the new orbital economy. It will be built upon atomic-level durability, and it’s time we ensure our hardware is truly ready for the long haul. Tony Morrin is the director of AMSCC Aerospace, which delivers flight-proven carbon composite gas tanks for satellites and launch vehicles that meet the exacting demands of modern space missions.

Blue Origin begins rebuilding New Glenn pad

Blue Origin has begun rebuilding its New Glenn launch pad after a severe explosion during a May 28 static-fire test caused significant damage, including a collapsed lightning tower. Despite the destruction, the company considers itself lucky that critical long-lead items like the propellant tank farm and a nearby booster were preserved. With cleanup and reconstruction underway, Blue Origin aims to resume launches by the end of the year.

WASHINGTON — Blue Origin has started rebuilding a launch pad severely damaged in a New Glenn explosion less than three weeks ago as it works to resume launches by the end of the year.

Speaking at the VivaTech conference in Paris on June 17, Blue Origin founder Jeff Bezos and its chief executive, Dave Limp, said they were making good progress on returning New Glenn to flight after a May 28 explosion during a static-fire test at Cape Canaveral's Launch Complex 36.

That explosion caused serious damage to the pad, including the collapse of a lightning tower and the destruction of the transporter-erector used to bring the rocket to the pad and raise it to the vertical position. However, Blue Origin concluded the explosion could have been worse.

It was a gut punch for the whole team. But what we've learned since then is that we got really lucky. Some of the long-lead items in the launch infrastructure were preserved.

That included a propellant tank farm at the pad, as well as a New Glenn booster used on the previous two launches that was in an integration hangar near the pad. "Various pieces of shrapnel missed the booster," he said. "There was a lot of good luck in that incident."

Limp said the company was able to call in a construction crew "just down the road" with 400 pieces of heavy equipment to work on cleaning up the pad. "The pad has been cleared of all debris. It's amazing how quickly that's happened," he said.

Just yesterday we started the reconstruction of the pad. We're going to fly this year.

"We'll fly before the end of the year," Bezos added.

Neither Bezos nor Limp discussed what caused the explosion or what changes the company will need to make to the vehicle or to pad infrastructure in response. Limp previously said the company would move to an "alternative vertical conop," or concept of operations, for installing the rocket on the pad, eliminating the need to replace the transporter-erector.

A relatively swift return to flight — many industry observers predicted it would take at least a year to rebuild the pad — would ease some of the pressure on a launch market where demand outstrips supply.

Demand for launch is insatiable right now. We have a tremendous backlog already on our books for launch. Every space launch company has tremendous backlog. We are supply-constrained. We are not demand-constrained.

That backlog includes missions for NASA as part of the agency's Artemis lunar exploration campaign using Blue Origin's Blue Moon lunar lander. The prospect of New Glenn being out of service for an extended time led NASA to consider "decoupling" Blue Moon from its New Glenn launcher.

Limp suggested that decoupling will not be necessary. "Just next year, early in the year, we'll fly our Mark 1 lander," he said, a reference to the

robotic Blue Moon Mark 1 lunar lander that will fly a mission NASA designated “Moon Base 1” last month.

That will be followed by the launch of a Blue Moon Mark 2 lander prototype in the middle of next year for the Artemis 3 mission, and another Mark 1 lander later in the year carrying NASA’s VIPER lunar rover to prospect for water ice in the south polar regions of the moon.

“The cycle time to the moon, the cadence to the moon, is going to increase very rapidly,” he said.

The company has continued work on the Blue Moon lander while recovering from the New Glenn explosion. That included a recent test of the BE-7 engine that powers the lander, during which the engine fired continuously for 41 minutes, the longest such test ever. The extended test, Bezos said, was needed because “the reliability of this engine is paramount.”

On social media, Limp posted a video of the full 41-minute test. “I’ve come to appreciate boring hotfires,” he wrote.

Academic

ACADEMIC · [spectrum.ieee.org] · [16 days ago] · 3,358 tokens / 0.0478c

7 Ways New Engineers Can Flourish in the Age of AI

New engineers entering the AI era must view artificial intelligence as leverage, not competition, to thrive. Success hinges on mastering fundamental programming concepts to understand underlying systems, even as AI assists with coding. Additionally, engineers need to learn how to effectively collaborate with AI by crafting clear prompts, reviewing generated code, and using it to boost productivity while maintaining critical judgment.

New graduates' careers are unfolding in an era when AI is not optional. The most successful engineers treat artificial intelligence as leverage, not competition. Here are seven tips to help keep young professionals in demand no matter how quickly the field's tools evolve.

1. Master the fundamentals first. AI tools can help you code, but you still need strong fundamentals in:
 - Data structures and algorithms for problem-solving.
 - [Operating systems], [databases], and networking for system-level understanding.
 - Core programming languages such as:
 - C++
 - Java
 - Python

AI can autocomplete syntax, but if you don't understand how things work under the hood, you're likely to struggle to debug or optimize.
2. Learn how to work with AI, not against it. The best engineers will not try to out-code AI. Instead, they will learn to:
 - Write clear prompts to generate better code snippets.
 - Review and debug AI-generated code for accuracy, performance, and security.
 - Use AI for productivity boosts while still exercising judgment.

Think of AI as a teammate. The real skill is knowing when to trust it and when not to.

3. Build projects that showcase end-to-end thinking. Employers increasingly look for engineers who can design and build systems, not just solve problems. Create projects that show you can:
 - Define requirements clearly.
 - Use AI tools responsibly within the workflow.
 - Deliver a product that scales and is maintainable.
4. Sharpen your system design skills early. Even junior engineers are now asked questions about basic system design with AI. Expect to explain to prospective employers:
 - How you would responsibly integrate AI into a system.
 - How to design fallbacks when AI fails.
 - How to ensure scalability and reliability.
5. Develop strong communication skills. Today's engineers don't just code in isolation. You will be expected to:
 - Explain design choices to teammates and stakeholders.
 - Document decisions clearly.
 - Collaborate effectively in cross-functional teams.

This is one area where AI cannot replace you. Clear communication is a career accelerant.
6. Stay curious and keep learning. The tech industry moves fast, and AI is accelerating that pace. Cultivate habits such as:
 - Following industry news, blogs, and open-source projects.

- Experimenting with new AI tools, frameworks, and libraries.
- Engaging in communities such as GitHub, IEEE Collabratec, LinkedIn, and Medium.

Employers value engineers who keep themselves sharp and relevant.

7. Think beyond coding. AI will increasingly handle routine coding tasks. The differentiators for you will be:
 - Problem-framing: Can you take a vague idea and turn it into a solution?
 - Architectural judgment: Can you design systems that scale and last?
 - Ethical awareness: Can you spot risks in AI use and address them responsibly?

Why Sardinians Are Fighting the Renewable Energy Transition

Sardinians are fiercely resisting the renewable energy transition, with a quarter of the voting population signing a petition to ban all new wind and solar projects on the island. This deep-seated "Not in my backyard" opposition, driven by local concerns, surprised an IEEE Spectrum editor reporting on energy storage there, highlighting the complex challenges of implementing green initiatives.

“Not in my backyard” is the rallying cry of citizens everywhere resisting projects proposed for their locality. Whether it’s affordable housing, a waste treatment plant, or a new data center, they may recognize the benefit of the activity. They just don’t want it near them. And the roots of that resistance differ from place to place. When it comes to the ongoing transition from fossil fuels to renewables, companies and policymakers need to know where, exactly, people are coming from.

The Italian island of Sardinia is a textbook example. As IEEE Spectrum’s power and energy editor Emily Waltz discovered when she traveled there last October, Sardinian opposition to wind and solar projects runs deep. It spurred a quarter of the voting population to queue up in public squares in 2024 to sign a petition banning all construction of renewable energy.

Waltz was surprised. She went there to see a promising new grid-scale energy storage system that uses domes inflated with carbon dioxide. While reporting on that project, she interviewed residents, engineers, activists, and professors about their attitudes toward climate change and the Italian government’s grand plans for renewable energy on the island. And Waltz soon learned of Sardinians’ profound antipathy toward renewable energy and its deep ties to a history of invasion, occupation, and exploitation stretching back 2,700 years.

It started with the Phoenicians and then extended through the Romans, the Byzantines, and the Iberians. Sardinia was absorbed into a newly unified Italy in 1861, and it became an autonomous region of Italy in 1948. The island’s population

is justifiably suspicious of outsiders, including the Italian government. “When you’re in Sardinia, the weight of history—you can feel it like in the air,” Waltz told me. “And it gets passed down from one generation to the next.”

Now, Italy needs Sardinia to produce even more power to meet the country’s climate goals—something that Sardinians see as Rome’s problem, not theirs. “Sardinia already exports about 30 percent of its electricity. It’s not like they need more,” Waltz says. “So it’s hard to make the case to build, build, build.”

The result of Waltz’s old-fashioned shoe leather reporting is this month’s cover story. She notes that the Sardinians she talked to aren’t climate-change deniers, and they don’t object to renewables per se. They just don’t like the way corporations and Italian policymakers are trying to plug into Sardinia like it’s one giant battery rather than the home of an ancient and proud people.

“I think Sardinians would be more receptive to renewable projects if it was more of a ground-up, grassroots approach,” Waltz says. Indeed, this homegrown approach is already working in some places in Sardinia. She knows of more than 50 projects, called energy communities, where the residents are deploying renewables themselves. The idea also holds promise for other places struggling to get locals to buy into the renewable-energy transition.

The Sardinian experience is both a cautionary tale and a blueprint. Ignore the weight of history that communities carry and your project risks failure. Meet the people where they are and you might just get somewhere. The same lesson applies

whether you're in Sulawesi or sub-Saharan Africa.
You just have to show up to learn it.

Artificial Intelligence

ARTIFICIAL INTELLIGENCE · [deepmind.google] · [1 month ago] · 1,786 tokens / 0.0229€

Finding the molecular switches behind new infectious diseases

Professor Clare Bryant is leveraging the AI tool Co-Scientist to identify the molecular switches that cause severe diseases when pathogens jump from animals to humans. This innovative tool has significantly accelerated her research by generating and refining hypotheses, even pinpointing specific amino acids for experimental focus, leading to novel insights and faster progress.

The majority of emerging infectious diseases are caused by pathogens that jump from animals to humans, such as Ebola, HIV, flu and Covid-19. Professor Clare Bryant at the University of Cambridge is using Co-Scientist to hunt for the molecular switches that cause severe diseases, like sepsis, in humans when pathogens leap between species, and find new approaches to prevent this happening.

Testing Co-Scientist, Bryant fed it a summary of one of her grant proposals studying flu in birds and humans outlining her lab's research questions. The tool generated and ranked a set of promising hypotheses — some she'd already considered, some she hadn't. The unfamiliar ones were the most thought-provoking.

When the grant was funded, Bryant fed in the full, detailed proposal. Later, reading through the output on a train to Brussels, she had an "a-ha!" moment: Co-Scientist had prioritised a protein that hadn't been on her radar, connected to several signalling pathways she was already interested in. She spent the rest of the week itching to give it more data.

Back at her lab, she added unpublished material, kept confidential within Co-Scientist. With each back-and-forth, the hypotheses sharpened, moving from candidate proteins down to the specific amino acids her lab could focus its experiments on.

Bryant's team is now building cell lines containing the amino acid mutations to test the refined hypotheses. To get to the point of identifying precise amino acids would normally have taken two to three years of experimental work. But her lab is now on track to complete it in six months, she says, if the work with Co-Scientist has led them to the right targets.

Programming

PROGRAMMING · [herbsutter.com] · [1.4 years ago] · 4,668 tokens / 0.0748c

New U.S. executive order on cybersecurity

The Biden administration's new cybersecurity executive order aligns with previous calls for significant improvements in programming language memory safety, especially within C++. While advocating for enhanced safety in C++'s key vulnerability areas, the author cautions that language safety alone is not a complete cybersecurity solution, as attackers will always seek alternative weaknesses.

The Biden administration just issued another executive order (EO) on hardening U.S. cybersecurity. This is all great stuff.

A lot of this EO is repeating the same things I urged in my essay nearly a year ago, “C++ safety — in context”... here's a cut-and-paste of my “Call(s) to action” conclusion section I published back then, and I think you'll see a heavy overlap with this week's new EO...

Call(s) to action

As an industry generally, we must make a major improvement in programming language memory safety — and we will.

In C++ specifically, we should first target the four key safety categories that are our perennial empirical attack points (type, bounds, initialization, and lifetime safety), and drive vulnerabilities in these four areas down to the noise for new/updated C++ code — and we can.

But we must also recognize that programming language safety is not a silver bullet to achieve cybersecurity and software safety. It's one battle (not even the biggest) in a long war: Whenever we harden one part of our systems and make that more expensive to attack, attackers always switch to the next slowest animal in the herd. Many of 2023's worst data breaches did not involve malware, but were caused by inadequately stored credentials (e.g., Kubernetes Secrets on public GitHub repos), misconfigured servers (e.g., DarkBeam, Kid Security), lack of testing, supply chain vulnerabilities, social engi-

neering, and other problems that are independent of programming languages. Apple's white paper about 2023's rise in cybercrime emphasizes improving the handling, not of program code, but of the data: “it's imperative that organizations consider limiting the amount of personal data they store in readable format while making a greater effort to protect the sensitive consumer data that they do store [including by using] end-to-end [E2E] encryption.” No matter what programming language we use, security hygiene is essential:

- Do use your language's static analyzers and sanitizers. Never pretend using static analyzers and sanitizers is unnecessary “because I'm using a safe language.” If you're using C++, Go, or Rust, then use those languages' supported analyzers and sanitizers. If you're a manager, don't allow your product to be shipped without using these tools. (Again: This doesn't mean running all sanitizers all the time; some sanitizers conflict and so can't be used at the same time, some are expensive and so should be used periodically, and some should be run only in testing and never in production including because their presence can create new security vulnerabilities.)
- Do keep all your tools updated. Regular patching is not just for iOS and Windows, but also for your compilers, libraries, and IDEs.
- Do secure your software supply chain. Do use package management for library dependencies. Do track a software bill of materials for your projects.

- Don't store secrets in code. (Or, for goodness' sake, on GitHub!)
- Do configure your servers correctly, especially public Internet-facing ones. (Turn authentication on! Change the default password!)
- Do keep non-public data encrypted, both when at rest (on disk) and when in motion (ideally E2E... and oppose proposed legislation that tries to neuter E2E encryption with 'backdoors only good guys will use' because there's no such thing).
- Do keep investing long-term in keeping your threat modeling current, so that you can stay adaptive as your adversaries keep trying different attack methods.

We need to improve software security and software safety across the industry, especially by improving programming language safety in C and C++, and in C++ a 98% improvement in the four most common problem areas is achievable in the medium term. But if we focus on programming language safety alone, we may find ourselves fighting yesterday's war and missing larger past and future security dangers that affect software written in any language.

Sadly, there are too many bad actors. For the foreseeable future, our software and data will continue to be under attack, written in any language and stored anywhere. But we can defend our programs and systems, and we will.

My main disappointment is that some of the provisions have deadlines that are too far away. Specifically: Why would it take until 2030 to migrate to TLS 1.3? It's not just more secure, it's also faster and has been published for seven years already... maybe I'm just not aware enough of TLS 1.3 adoptability issues though, as I'm not a TLS expert.

Here in the United States, we'll have to see whether the incoming administration will continue this EO, or amend/replace/countermand it. In the United States, that's a drawback of using an EO compared to passing an actual law with Congressional approval... an EO is "quick" because the President can issue it without getting legislative approval (for things that are in the Presidential remit), but for the same reason an EO also isn't "durable" or guaranteed to outlive its administration. Because

the next President can just order something different, an EO's default shelf life is just 1-4 years.

So far, all the major U.S. cybersecurity EOs that could affect C++ have been issued since 2021, which means so far they have all come from one President... and so we're all going to learn a lot this year, one way or another, about their permanence. (In both the U.S. and the E.U., actual laws are also in progress to shift software liability from consumer to software producers, and those will have real teeth. But here we're talking about the U.S. EOs from 2021 to date.)

That said, what I see in these EOs is common sense pragmatism that's forcing the software industry to eat our vegetables, so I'm cautiously optimistic that we'll continue to maintain something like these EOs and build on them further as we continue to work hard to secure the infrastructure that our comfortable free lifestyle (and, possibly someday, our lives) depends on. This isn't about whether we love a given programming language, it's about how we can achieve the greatest hardening at the greatest possible scale for our civilization's infrastructure, and for those of us whose remit includes the C++ language that means doing everything we can to harden as much of the existing C and C++ code out there as possible — all the programmers in the world can only write so much new/rewritten code every year, and for us in C++ by far the maximum contribution we can make to overall security issues related to programming languages (i.e., the subset of security issues that fall into our remit) is to find ways to improve existing C and C++ code with no manual source code changes — that won't always be possible, but where it's possible it will maximize our effectiveness in improving security at enormous scale.

Speaking at New York C++ meetup on January 13

The author will be speaking at the New York C++ meetup on January 13 in Midtown East, presenting a condensed update of their "Peering forward: C++'s next decade" talk with ample time for Q&A. Sponsored by Citadel Securities, the free event includes food and drink, but preregistration RSVP is required to attend.

Less than two weeks from now, on January 13 I'll be speaking at the New York C++ meetup in Midtown East (Clinton Hall at 230 E 51st Street). I'll be giving a condensed update of my recent "Peering forward: C++'s next decade" talk, so that there'll be plenty of time for Q&A — please have your questions ready about all the cool things happening right now in the ISO C++ world.

The meetup is sponsored by Citadel Securities, where I work. Food and drink will be served! Registration is free, but preregistration RSVP is required so please use the form at the link about to reserve your spot. A waitlist is available if space runs out.

I hope to see many of you there!

Math

MATH · [terrytao.wordpress.com] · [3 days ago] · 5,204 tokens / 0.0884c

Third SAIR competition: inverse Galois challenge

The third SAIR challenge has been launched, focusing on generating numerical data for the inverse Galois problem. This collaborative project with the L-functions and modular forms database (LMFDB) aims to complete a large, verified mathematical dataset, with verification performed by MAGMA and PARI/GP.

I am happy to announce the third SAIR challenge, which is focused on obtaining numerical data for the infamous inverse Galois problem. This is a collaborative project with the L-functions and modular forms database (LMFDB), and is organized by John Jones, Jen Paulhus, David Roe, Andrew Sutherland, and myself. The challenge is somewhat similar to my own Equational Theories Project, in that one is trying to complete a large mathematical data set in a verified fashion, except that the target data set had an existing mathematical interest. Also, the verification will be done by MAGMA (as well as PARI/GP) rather than Lean.

Let me first quickly review the inverse Galois problem. Suppose one has an irreducible polynomial of one variable of some degree and integer coefficients; take for instance $x^3 - x - 1$. Then $P(x)$ will have n distinct roots $r_1, \dots, r_n$; in this case the roots happen to be r_1, r_2, r_3 . The roots generate a splitting field $K = \mathbb{Q}(r_1, \dots, r_n)$ over the rational numbers $\mathbb{Q}$. Any automorphism of this splitting field must permute the roots $r_1, \dots, r_n$, and thus generates a subgroup of the permutation group S_n (defined up to relabeling of the roots), which we call the Galois group of P . This is some subgroup of S_n that acts transitively on the n roots (because each root generates the field). Typically, it is all of S_n ; but occasionally it is smaller. For example, the particular cubic polynomial $x^3 - x - 1$ above has the special property that each root r_i individually generates the entire field K .

, thanks to the identities $r_2 = r_1^2 - 1$ and $r_3 = -r_1^2 - r_1 + 1$. Because of this, the Galois group of $x^3 - x - 1$ is the cyclic group C_3 (or equivalently, the alternating group A_3), rather than the full symmetric group S_3 . (This is in contrast to, say, $x^3 - 2$, whose roots $\sqrt[3]{2}, \sqrt[3]{2}\omega, \sqrt[3]{2}\omega^2$ cannot be expressed as rational polynomials of each other, and whose Galois group is all of S_3 .) In fact, in the cubic case, it turns out that the Galois group is C_3 when the discriminant is a perfect square, and S_3 otherwise.

More generally, we have

Problem 1 (Inverse Galois Problem)

Let G be a transitive permutation group on n letters. Can G be realized as the Galois group of some degree n irreducible polynomial $P(x)$ with integer coefficients (after identifying the n roots of P suitably with the n letters)?

The answer to this problem is known to be positive for $n \leq 23$, with the single possible exception of the sporadic Mathieu group M_{23} : there are 25000 transitive permutation groups on $n \leq 23$ letters (cf. OEIS A002106), and for 24999 of them, a polynomial has been located with that Galois group; see this database of Klüners and Malle. The problem of locating a polynomial with Galois group M_{23} is a notorious open problem, though this is likely to be quite a difficult problem, and not the objective of the SAIR challenge.

Instead, we will focus on “breadth” rather than “depth”, in order to leverage the power of crowdsourcing and modern AI technologies. It turns out that there are 25000 distinct transitive permutation groups on $n \leq 23$ letters, which are conventionally labeled from T_1 (the cyclic group C_n) to T_k (the permutation group S_n). The first stage of the challenge will be:

Problem 2 (First stage of SAIR challenge)

For as many of the groups T_k , $n \leq 23$, locate an integer polynomial with that Galois group (up to isomorphism). (Also of interest is to specify the number of real roots, and to keep the discriminant low; more on this later.)

The verification side of this problem is essentially solved: the MAGMA computer algebra system can take any candidate polynomial and locate its Galois group within seconds. The MAGMA team has kindly granted SAIR a limited license to provide an API for contestants to calculate a certain number of Galois groups per day without needing to purchase their own license, though of course they are free to use their other computational tools to also perform these calculations outside of the competition.

The LMFDB already has polynomials for 286 of the 25000 groups, so there is plenty of remaining polynomials to claim in the challenge.

For applications, it is of interest to track some other statistics of a polynomial besides its Galois group. One of these is the number r of real roots, which is a number between 0 and n of the same parity as n (and which has to be achievable as the number of fixed points of one of the permutations in the Galois group, namely the one corresponding to complex conjugation); in particular, this number must be even in the degree n case. Combining the label T_k of the Galois group with the number of roots r turns out to generate 25000 $\times n$ pairs in degree n , and the challenge is actually to attach polynomials to as many of these pairs as possible. (The LMFDB has already done so for just 1000 of these.)

Of course, there are infinitely many polynomials of degree n , and any Galois group that is representable by one polynomial, will be representable by infinitely many others (e.g., one could simply trans-

late the polynomial by an arbitrary integer shift). To avoid creating an unusable database filled with uninteresting polynomials, we will prioritize polynomials whose (absolute) discriminant is as small as possible. (There are some technical details as to how this discriminant is defined and computed; see this page for details). The way we have set things up, each (T_k, r) pair will come with a leaderboard for the polynomials with the smallest discriminants that have been located so far by contestants, removing duplicates arising from trivial operations such as translating the polynomial. Contestant team will be awarded a score between 0 and 1 for each submitted polynomial based on how small their discriminant is compared to the best known discriminant, and how many other teams were also able to find a polynomial with that (T_k, r) pair. Thus, pairs that are extremely easy to generate (such as those associated to the full permutation group S_n) will be worth only a negligible score (as every contestant will be able to submit a polynomial for that pair), while pairs which are difficult to locate a polynomial for will be worth more points.

For this competition, the unrestricted use of any sort of computational tool, including AI, to locate the polynomials, are expressly permitted; this first stage of the competition is a “black box” challenge where we are not directly interested in obtaining insights as to how the polynomials are located, but the sole objective is to resolve as much of the inverse Galois challenge as possible. As such, the notorious uninterpretability of modern AI is not a concern for this stage. However, we will encourage contestants to share techniques with each other in order to cover more ground, through the Zulip channel for this challenge.

This first stage of the competition will close on August 15. After this, we will launch a second stage (with details to be determined) to focus on some set of candidate Galois groups that could not be resolved by the first stage. Here we envisage a more collaborative, conceptual, and human-driven effort in which the role of AI tools may be more secondary, and with more of a focus on creating mathematically interesting results rather than simply trying

to saturate a given benchmark. Stay tuned for more details!

Science

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Why the Human Genome's Tangled Physicality May Confound AI

Despite the human genome's full sequencing, its intricate physicality and the vast non-coding regions that regulate gene activity present a complex puzzle, challenging the initial view of DNA as a simple blueprint. Understanding how genes are switched on and off, rather than just their sequence, is now the critical question, potentially confounding AI's ability to fully decipher life's secrets.

Since its molecular structure was deduced in the 1950s, DNA has been hailed by many biologists as the secret of life. They've read and studied the information stored in the DNA found in the cells of living organisms, known as their genomes, and claimed that this genetic database must be some kind of blueprint, code script, or computer. But if DNA really does harbor some greater secret about how life works, biologists have yet to find it.

In fact, the human genome is less a script than a puzzle that gets harder the closer they look. Knowing the entire sequence — the order of all 3 billion or so of our DNA's chemical building blocks, nearly fully deduced by the international Human Genome Project between 1990 and 2003 — hasn't helped much. That investigation showed that barely 2% of the human genome consists of actual genes, the information-coding sequences of DNA.

It's now clear that understanding the human genome is no longer a matter of figuring out what each gene does. The deeper and much harder question is how those genes are used, or regulated, a question that seems to involve some and perhaps much of the rest of the genome. By switching suites of genes on and off, the many different cell types in our bodies can all be created from the same material. Cells also regulate their genes from moment to moment in response to a constant inflow of signals from their neighbors and surroundings. But the processes that govern gene regulation are proving so complex that some biologists wonder whether a full understanding of it — of how the genome really

works — will ever be within the grasp of our puny minds.

Some are counting on outsourcing the analysis to artificial intelligence. Genomic “foundation models” such as Evo 2, Genos, and Google DeepMind's AlphaGenome are trained on vast quantities of genomic data, which biologists use to make predictions about how differences in DNA sequence affect biological processes and ultimately the traits (including disease risk) of a whole organism. These algorithms don't worry about the complicated regulatory stuff going on; all of that is supposedly subsumed by the algorithm's “training,” through which it deduces correlations from cases we already know about.

This approach is likely to be useful, but for those who crave real understanding of how the genome, and ultimately life itself, works, a computational black box will never suffice. And perhaps more to the point, the genome might not submit to the kind of straightforward input-output approach that such AI models ultimately assume.

That's because the genome is no blueprint or algorithm. It is something else.

The Old View

Given that it's the product of around 4 billion years of evolution, perhaps it's not surprising that our genome is complicated. The surprise has been what those complications are. “Our genome is not what we might make it if we sat down at the drawing board,” said the biologist Karen Adelman, who studies gene regulation at Harvard Medical School.

The traditional view posits that a small proportion of our DNA holds the code for making the protein molecules that orchestrate our cells' chemistry. Each instruction for a protein is held in a corresponding gene — we have around 20,000 of these — and gene sequences can range in length from a couple of dozen to almost 3 million DNA “letters” (representing molecules called nucleotides). Making a protein from its gene is a two-stage affair. First the DNA is read, letter by letter, by an enzyme called a polymerase, which creates a copy of that code in a related molecule called messenger RNA (mRNA). This is called transcription. The mRNA is then read by a piece of molecular machinery called the ribosome, which constructs the protein — a process called translation. The proteins made by the ribosome then go off to do their jobs in making and sustaining the organism.

This picture is still more or less correct. But it turns out that “the genes are probably not the most interesting part of the genome,” Adelman said.

What matters more is how our genes, many of which we share with simpler organisms, are regulated: turned on and off. Which proteins a cell needs changes over time and according to cell type: muscle, brain, skin, and so on. How the genes that encode those proteins are regulated depends on some of the genome that doesn't code for proteins.

Biologists have known about gene regulation, and the involvement of “noncoding” DNA, since the 1960s. But for many years, most of what they understood about this came from studies of simple organisms like bacteria, where the principles are generally straightforward. It has gradually become clear, though, that in complex eukaryotic organisms like us, gene regulation is far more complicated, involving overlapping systems of oversight and control, each with its own intricacies.

Transcription Factors

Transcription gets started by proteins called transcription factors, which are like the operations managers of gene regulation. These proteins stick to sections of DNA (typically close to the target gene) and recruit the polymerase enzyme to make an mRNA copy. In bacteria, transcription factors are rather like keys that fit the locks of unique binding

sites on DNA. But that's not how they work in complex organisms. In us, the logic of transcription factors is more difficult to parse.

For one thing, our transcription factors don't show strong preferences for particular DNA binding sites. What's more, they tend to work in pairs or groups. And a given transcription factor might have different effects in different contexts, such as activating gene transcription in one cell type but suppressing it in another, depending on which other transcription factors are around.

In bacteria, regulation tends to have an “OR” logic, Adelman said, whereby a particular signal turns a gene on or off: It's either this or that. But in the human genome the logic is more like what computer scientists designate “AND.” Many signals are integrated to reach a regulatory decision: this and that and also that other thing. In this case, regulation can be more responsive to nuances of context, and the regulatory knobs are tunable rather than being just on/off. “This is part of the beauty” of our regulatory complexity, Adelman said.

When they interact with the genome, transcription factors bind to pieces of DNA called enhancers — which present a puzzle of their own.

Enhancers

Enhancers are gathering points for transcription factors, and they are thought to be the decisive influence on transcription: They deliver the “go” signal for a waiting polymerase to make an mRNA version of the DNA sequence. Seems simple enough, but mapping enhancers to their respective genes is far from straightforward. Our genome has hundreds of thousands, perhaps millions, of enhancers. That means we have many more of them than we have genes. Each gene might be influenced by many enhancers, and each enhancer might influence multiple genes.

“It's embarrassing that 25 years after the Human Genome Project, we don't know where all the enhancers are in the genome, let alone what they do when they act and which genes they control,” said Wendy Bickmore, a genome biologist at the University of Edinburgh.

Biologists do know that most enhancers won't respond to a single transcription factor. Their acti-

vation “requires a cocktail,” Bickmore said. “That’s what gives [an enhancer] that exquisite specificity — because it’s only in a particular cell at a particular time that you have the right combination of factors to bind and activate that enhancer.”

Some enhancers are, as you’d expect, close to the genes they regulate, or even sit on DNA inside a gene. But others sit far away from the gene — perhaps millions of nucleotides away, with more genes in between.

The existence of such so-called “distal” enhancers “seems bonkers,” Bickmore said. “How do you get that information from over there to over here, to the gene that needs to be activated? That’s a largely unanswered question.”

One of the answers comes in the form of a loop.

Loops and Hubs

Distal enhancers are brought to the gene they regulate on great loops of DNA or, more strictly, of chromatin, the combination of DNA and its packaging proteins that are unraveled as if from a ball of wool. The loops are created by a protein motor called cohesin, which runs up and down the DNA strand and extrudes it as needed.

Once cohesin has formed a loop to bring elements together, what then? It was once thought that they then stick together or assemble into a molecular machine, but they don’t. Rather, the components appear to form a loose but dense blob in which they interact rather weakly, fleetingly, and indiscriminately — a sort of committee, sometimes called a condensate.

These transcription hubs are extremely fluid and differ from one cell to another. “There’ll be a bit of loop extrusion going on over here, in the next cell it might be over here, and the whole thing is turning over incredibly fast,” Bickmore said. Even if the cells are notionally identical — both skin cells, say — exactly what the gene-regulatory machinery is up to at any moment is never quite the same in any two of them.

Chromatin loops are just one reason why a gene’s transcription depends on the shape and structure of the chromatin around it.

Chromatin Shape

The textbook image of a chromosome — one of the 46 units into which our genomes are divided — is of a compact, X-shaped cluster of chromatin. But any time a cell is not actively dividing, its chromatin is unwound into what looks like a tangled mess. There is order to the chaos, however. Some parts of chromatin are densely packed into a form called heterochromatin. The compacted DNA there is relatively inaccessible to transcription factors; the genes it contains are typically silenced. Meanwhile, other parts are relatively loose, open, and accessible: This is called euchromatin.

There are special enzymes involved in packaging and repackaging chromatin, thereby controlling transcription. In other words, what matters is not just the encoded information in the DNA but also how it exists physically and dynamically in space. “We’ve stopped thinking about the genome as a linear piece of DNA code,” Bickmore said. “Thinking about this incredibly dynamic three-dimensional folding as absolutely inherent to regulation is a very exciting change.”

One aspect of this 3D organization is the clustering of segments of chromatin into compartments called topologically associating domains (TADs). Within a TAD, the genes seem to be coregulated: switched on or off in groups. Such groups keep suites of genes active or silent together to form and provide function in different cell types. Cohesin is also involved in the shuffling of chromatin to construct TADs — a dynamic process in which the chromatin is constantly rearranged in our cells.

Chromatin shape can also be influenced by chemical modifications called epigenetic marks: small molecules attached to DNA packaging proteins called histones or stuck directly to DNA. Some of these epigenetic modifications can alter the electrical charges on histones, which changes how the proteins attract or repel one another and so rejigs the chromatin packing. Epigenetic modifications to chromatin are like annotations of the DNA script that change its meaning in a given context. When cells divide, the epigenetic annotations are copied, too.

How and when the marks get added and changed, and what each type of mark means for gene activity,

are complex questions with no simple answers. Some researchers talk of an “epigenetic code” governing this aspect of gene regulation, but it’s far from clear if anything so systematic really exists.

All of these processes and others can determine whether a gene gets transcribed into mRNA. But there are further layers of regulation that determine whether the mRNA is then translated into a corresponding protein — and which protein arises.

RNA Interventions

This post-transcriptional regulation is often controlled by RNA molecules that are said to be noncoding. These short-lived molecules aren’t templates for proteins, as mRNA is, but have other jobs of their own. While mRNA is produced from the protein-coding areas of DNA (so-called “coding genes”), noncoding RNAs are transcribed from other DNA regions now generally described as non-coding genes. These noncoding RNAs are versatile, taking on varied roles in a cell. Researchers are learning more about what they can do every day, and many if not most of them seem to be involved in gene regulation.

Small noncoding RNAs called microRNAs, for example, can silence mRNAs before they can be translated into proteins. They do this by guiding special enzymes to a particular mRNA to degrade or chemically modify it. The microRNAs don’t do this job alone but, not unlike transcription factors, act combinatorially, in groups, and in a rather promiscuous manner: A given microRNA might regulate many mRNAs, and a given mRNA might be regulated by many microRNAs.

Why make an mRNA only to stop it getting translated in a protein? This sort of post-transcriptional regulation is like having another checkpoint: Does the cell really need this protein? MicroRNAs can be mobilized to allow cells to adjust gene expression depending on the immediate context. In this way, the workings of the genome are less like a program’s inevitable progression and more like an adaptive and responsive process.

Another post-transcriptional complication is that mRNAs get translated to protein only after they have been reorganized. Fresh from transcription, an mRNA contains sequences that encode bits of

protein, called exons, as well as sequences that shouldn’t be translated and need to be snipped out, called introns. (Strictly speaking, this pre-edited RNA is called pre-mRNA.) The job of editing introns out and splicing exons together is done by a molecular assembly called the spliceosome, which is made from several proteins together with various noncoding RNAs.

The spliceosome too can be sensitive to context, so that it might splice the pre-mRNA to encode one protein in one cell type and a slightly different protein in another. Sometimes these different protein “isoforms” can have very different roles. Transcription factors, for example, are often alternatively spliced in this way, and their isoforms can take on different regulatory tasks — some might activate gene expression, for instance, while others repress it.

Checks and Balances

All told, these and other regulatory mechanisms show that the genome is far from some automated program running in the background to build us and keep us alive. Our cells are, in effect, making complex decisions about how to use their genes — both the information they contain and the structure they assume.

Thus, cells need to assemble a rather loose and fuzzy committee of components, such as transcription factors and enhancers, to get transcription underway, which also depends on how the chromatin strand is shaped and molded at that moment. Then there are further layers of decision-making and action-taking in between mRNA and the final, functional protein.

Remember, too, that all the players — from transcription factors to noncoding RNAs — are themselves produced from the genome in the same kind of context-dependent process. That makes the genome rather like a recursive, self-referential system that the computer scientist Douglas Hofstadter dubbed “a strange loop.” It acts on itself, mindful of its own history (which determines chromatin conformation and epigenetic markings, say) and heedful of messages from inside and outside the cell. Not, then, a blueprint.

And for that reason, not at all easy to understand. “I wouldn’t have designed it this way if I was God,” Bickmore said. “But here we are!”

Why is gene regulation in animals like us so darned complicated? One potential answer is that evolution doesn’t have the foresight to design with efficiency and transparent logic, but merely tinkers with what it has already available. Maybe so — but eukaryotic gene regulation isn’t just a messy version of what happens in bacteria. It has different principles, and there’s surely a reason for them.

Bickmore suspects that the complexity of regulation and of genome organization might have been the only means of generating complexity in the organism. For example, organisms with many tissue types and varied lifestyles required more control over which genes were on or off in a given cell. One thing this demanded was more and more noncoding regulatory sequences in DNA. But then they couldn’t all fit close to the gene itself.

“As you get more complexity, you need to add more and more enhancers,” Bickmore said. “But where are you going to put them? You start to put them farther and farther away. Once they are [far enough], you start to need TADs and three-dimensional [chromatin] folding to allow those things to work.”

We also need regulatory complexity because, over evolutionary time, the human genome has accumulated DNA from parasitic viruses in the form of jumping genetic material called transposable elements. These sequences have inserted themselves all over our chromosomes and are good at replicating themselves. To sift the good DNA from the bad, we needed additional layers of regulation to ensure that cells weren’t translating RNAs they don’t really need or that could be actively harmful.

With so many context-dependent checks and balances in the workings of our genome, it is evidently not a program or algorithm that predictably generates the same outcome in every situation. It’s an open informational system that responds to external inputs and the genome’s dynamic internal conditions. This poses a challenge if AI relies solely on the genetic sequences within genomes to predict what genomes will do.

A Highly Sensitive Organ

Researchers developing AI-based genomic foundation models such as AlphaGenome hope that all these layers of regulation — transcription factors, splicing, epigenetic marks, loops, chromatin packing, and so on — will be implicitly included in the correlations that the algorithms learn between genetic sequence and organismal traits. They’re content for the complexity described above to be in a black box, so long as the model generates accurate predictions. But will that work?

“I’m sure [AlphaGenome] is going to be useful, but with limitations,” Bickmore said. “To me the big gap is in the complexity of the human body — in all the cell types and how they change over time in development. And all that data is missing.”

Fundamentally, the challenge is that the genome is not a set of static, linear instructions. It is highly dynamic, and it uses its information contextually, with combinatorial and promiscuous logic. “Whether we’ll ever be able to capture that aspect” in algorithms like AlphaGenome, “I don’t know,” she said.

Yet the problem goes even deeper because the functioning of specific organisms, including each of us, doesn’t just depend on genomes. Other factors, such as diet, environment, microbiome and, for us at least, culture, can matter hugely, too — not just in terms of how we act and how healthy we are but also in the state of our genome itself. The biologist Adrian Woolfson, co-founder of California-based biotech company Genyro, which aims to use AI systems for so-called “generative biology,” calls this information cloud the “informiome.”

“While the human genome forms the foundation of the human informiome, other layers of extra-genetic information are equally important,” Woolfson wrote in his book *On the Future of Species*, published in April 2026. Genomic foundation models won’t even be able to predict all the consequences of genetic mutations, he argued, because the relevant information is not in the genome sequence in the first place.

So how should we think about the genome? Maybe the only metaphors that can capture the way the genome really works must come from biology itself. In 2020, the biological historian Evelyn Fox

compared the genome to “an exquisitely sensitive reactive system.” Rather than a sequence of genes leading to the formation of traits, she said, it’s more of “a device for regulating the production of specific proteins in response to constantly changing signals it receives from its environment.”

That sounds close to the picture painted by the geneticist Barbara McClintock in the address she delivered upon being awarded the 1983 Nobel Prize in Physiology or Medicine for her discovery of transposons. The genome, she declared, is

a highly sensitive organ of the cell, monitoring genomic activities and correcting common errors, sensing the unusual and unexpected events and responding to them, often by restructuring the genome.

Research since that time has fleshed out this image, revealing how the shape of chromatin can matter as much as the information its DNA sequences encode and how an army of molecules collaborates to reorganize it and make collective decisions about how to use its genetic information in context-dependent ways. There is no human technology that works this way, so metaphors such as blueprints, programs, or computers will always fall short.

Bickmore is optimistic that the workings of the genome are understandable, despite its complexity. “We’ve got a handle on it now,” she said. “We might not know the details, but I think the whole field is coalescing now into a framework where we’re thinking along similar lines.” AI can surely help with this sense-making, but in the end, human reasoning will be needed to discern the fundamental principles.

“McClintock was far more on point than people realized at the time,” Adelman said. “What she said was that the genome isn’t static — it’s living.”

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Epson HX-20 Gets A Drive Upgrade

Andrew Menadue has ingeniously upgraded the vintage Epson HX-20, replacing its unreliable microcassette drive with a modern solution. Utilizing a Raspberry Pi Pico and microSD card, his device emulates the original tape drive and can even mimic ROM cartridges, ensuring the classic computer's longevity.

The Epson HX-20 is sometimes referred to as an early laptop computer. It's a little odd in its form factor, and in its storage, relying on a microcassette drive to store data. It can be problematic to keep these tapes and drives going after so many decades, so Andrew Menadue has been tinkering with a more modern solution.

The replacement drive uses a Raspberry Pi Pico to emulate the original tape drive. The Pico uses a microSD card to store data instead of the magnetic media of old. The device has a small screen for showing status information and four buttons for navigation, allowing the faux drive to be controlled as to what "tape" it's pretending to be. It's also possible to use the device to emulate ROM cartridges that could be used with the HX-20 in place of its original tape deck storage solution.

We've seen some other old hardware get similar drive upgrades before, too. No surprise, because mechanical drives and media simply don't last forever. Sometimes you need to build a replacement that's viable today. Video after the break.